

Homework problemsSOLUTIONS

- ① Solve $u_t - Ku_{xx} = 0$ for $u(x, t)$, $0 \leq x \leq l$, with boundary conditions $u(0, t) = u(l, t) = 0$, and initial condition $u(x, 0) = T$.

Using variable separation, seek solution in the form:

$$u(x, t) = v(x) q(t).$$

Substituting into the heat equation, we have:

$$v(x) \frac{dq}{dt} - q(t) K \frac{d^2 v}{dx^2} = 0.$$

Dividing by $v(x) q(t)$:

$$\underbrace{\frac{1}{q(t)} \frac{dq}{dt}}_{-\lambda} - K \underbrace{\frac{1}{v(x)} \frac{d^2 v}{dx^2}}_{-\lambda} = 0$$

Solving the time equation,

$$\frac{dq}{dt} = -\lambda q,$$

we obtain $q(t) = Ce^{-\lambda t}$.

The coordinate equation:

$$\frac{K}{v(x)} v''(x) = -\lambda,$$

$$v''(x) + \frac{\lambda}{K} v(x) = 0.$$

The general solution of this equation is

$$v(x) = A \cos \sqrt{\frac{\lambda}{K}} x + B \sin \sqrt{\frac{\lambda}{K}} x.$$

To satisfy the boundary conditions, we require

(2)

$$v(0) = v(l) = 0.$$

For $x=0$, we have $A=0$,

hence
$$v(x) = B \sin \sqrt{\frac{\lambda}{K}} x.$$

For $x=l$:

$$B \sin \sqrt{\frac{\lambda}{K}} l = 0,$$

so that
$$\sqrt{\frac{\lambda}{K}} l = n\pi, \quad n=1, 2, \dots$$

$$\sqrt{\frac{\lambda}{K}} = \frac{n\pi}{l}$$

$$\lambda = \frac{n^2 \pi^2}{l^2} K.$$

Hence:

$$u(x,t) = B_n \sin \frac{n\pi x}{l} e^{-\frac{n^2 \pi^2}{l^2} K t},$$

where B_n is an arbitrary constant.

Combining these solutions using the superposition principle,

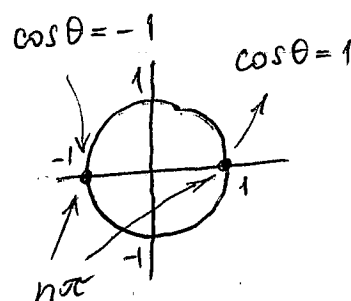
$$u(x,t) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l} e^{-\frac{n^2 \pi^2}{l^2} K t} \quad (*)$$

To satisfy the initial conditions at $t=0$, we require

$$T = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l}.$$

Finding B_n :

$$\begin{aligned} B_n &= \frac{2}{l} \int_0^l T \sin \frac{n\pi x}{l} dx \\ &= \frac{2T}{l} \frac{l}{n\pi} \left(-\cos \frac{n\pi x}{l} \right) \Big|_0^l \\ &= \frac{2T}{n\pi} \left(-\underbrace{\cos n\pi}_{(-1)^n} + 1 \right) \\ &= \frac{2T}{n\pi} (1 - (-1)^n) \end{aligned}$$



Hence:

$$B_n = \begin{cases} 0, & n = 2m \quad (\text{even } n) \\ \frac{4T}{n\pi}, & n = 2m+1 \quad (\text{odd } n) \end{cases} \quad m = 0, 1, 2, \dots \quad (3)$$

Using this in (*), we obtain the final answer:

$$u(x,t) = \frac{4T}{\pi} \sum_{m=0}^{\infty} \frac{\sin \frac{(2m+1)\pi x}{l}}{2m+1} e^{-\frac{(2m+1)^2 \pi^2}{l^2} Kt}.$$

② Solving the same problem as ① with a different initial condition, $u(x,0) = \frac{4T}{l^2} x(l-x)$, we repeat all steps that led to equation (*) (page 2). Imposing the initial condition:

$$\frac{4T}{l^2} x(l-x) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l}$$

$$B_n = \frac{2}{l} \int_0^l \frac{4T}{l^2} x(l-x) \sin \frac{n\pi x}{l} dx$$

$$= \frac{8T}{l^3} \left[l \int_0^l x \sin \frac{n\pi x}{l} dx - \int_0^l x^2 \sin \frac{n\pi x}{l} dx \right]$$

$$= \frac{8T}{l^3} \left[lx \frac{l}{n\pi} \left(-\cos \frac{n\pi x}{l} \right) \Big|_0^l + \frac{l^2}{n\pi} \int_0^l \cos \frac{n\pi x}{l} dx \right] \quad \left. \begin{array}{l} \text{Integrating} \\ \text{by parts} \end{array} \right\}$$

$$+ x^2 \frac{l}{n\pi} \cos \frac{n\pi x}{l} \Big|_0^l - \frac{2l}{n\pi} \int_0^l \cos \frac{n\pi x}{l} x dx \Big]$$

$$= \frac{8T}{l^3} \left[-\frac{l^3}{n\pi} \cos n\pi + \frac{l^2}{n\pi} \cdot \frac{l}{n\pi} \sin \frac{n\pi x}{l} \Big|_0^l + \frac{l^3}{n\pi} \cos n\pi - \frac{2l}{n\pi} \frac{l}{n\pi} \sin \frac{n\pi x}{l} \Big|_0^l + \frac{2l^2}{n^2 \pi^2} \int_0^l \sin \frac{n\pi x}{l} dx \right] \quad \left. \begin{array}{l} \sin n\pi = 0 \\ \sin 0 = 0 \end{array} \right\}$$

$$= \frac{8T}{l^3} \left[\frac{2l^2}{n^2\pi^2} \frac{l}{n\pi} \left(-\cos \frac{n\pi x}{l} \right) \right]_0^l$$

$$= \frac{16T}{n^3\pi^3} (1 - (-1)^n) = \begin{cases} 0, & n = 2m \\ \frac{32T}{n^3\pi^3}, & n = 2m+1, \\ & m = 0, 1, 2, \dots \end{cases}$$

Substituting this into (*) we obtain the answer:

$$u(x,t) = \frac{32T}{\pi^3} \sum_{m=0}^{\infty} \frac{\sin \frac{(2m+1)\pi x}{l}}{(2m+1)^3} e^{-\frac{(2m+1)^2\pi^2}{l^2} Kt}.$$

③ Heat equation: $u_t - Ku_{xx} = 0$, $0 \leq x \leq l$
 boundary conditions: $u_x(0,t) = u_x(l,t) = 0$,
 initial condition: $u(x,0) = \frac{4T}{l^2} x(l-x)$.

Using variable separation:

$$u(x,t) = v(x)q(t),$$

$$v(x) \frac{dq}{dt} - q(t) K \frac{d^2v}{dx^2} = 0$$

$$\underbrace{\frac{1}{q(t)} \frac{dq}{dt}}_{-\lambda} - \underbrace{\frac{K}{v(x)} \frac{d^2v}{dx^2}}_{-\lambda} = 0$$

Time equation $\frac{dq}{dt} = -\lambda q \Rightarrow q(t) = Ce^{-\lambda t}$.

Coordinate equation:

$$\frac{d^2v}{dx^2} + \frac{\lambda}{K} v(x) = 0$$

$$\Rightarrow v(x) = A \cos \sqrt{\frac{\lambda}{K}} x + B \sin \sqrt{\frac{\lambda}{K}} x.$$

Boundary condition on $u_x(x,t) = v'(x)q(t)$,

requires

$$v'(0) = 0, \quad v'(l) = 0.$$

(5)

$$v'(x) = -A \sqrt{\frac{\lambda}{K}} \sin \sqrt{\frac{\lambda}{K}} x + B \sqrt{\frac{\lambda}{K}} \cos \sqrt{\frac{\lambda}{K}} x.$$

$$x=0: \quad B \sqrt{\frac{\lambda}{K}} \cdot 1 = 0 \Rightarrow B = 0 \quad (\text{or } \lambda = 0 \text{ see below})$$

$$\Rightarrow v(x) = A \cos \sqrt{\frac{\lambda}{K}} x.$$

$$x=l: \quad v'(l) = 0 \quad \text{gives} \quad -A \sqrt{\frac{\lambda}{K}} \sin \sqrt{\frac{\lambda}{K}} l = 0$$

$$\Rightarrow \sqrt{\frac{\lambda}{K}} l = n\pi, \quad n = 0, 1, 2, \dots$$

$$\sqrt{\frac{\lambda}{K}} = \frac{n\pi}{l}$$

$$\lambda = \frac{n^2 \pi^2}{l^2} K$$

$$\text{Hence:} \quad u(x,t) = A_n \cos \frac{n\pi x}{l} e^{-\frac{n^2 \pi^2}{l^2} K t},$$

where A_n are arbitrary constants.

Using the superposition principle, we construct:

$$u(x,t) = \sum_{n=0}^{\infty} A_n \cos \frac{n\pi x}{l} e^{-\frac{n^2 \pi^2}{l^2} K t} \quad (1)$$

This solution must satisfy the initial condition.

Setting $t=0$:

$$\frac{4T}{l^2} x(l-x) = \sum_{n=0}^{\infty} A_n \cos \frac{n\pi x}{l} \quad (2)$$

The right-hand side is a cosine Fourier series, and traditionally, the $n=0$ term is written out separately, so that (1) becomes (writing $\frac{A_0}{2}$ instead of A_0)

$$u(x,t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi x}{l} e^{-\frac{n^2 \pi^2}{l^2} K t}, \quad (3)$$

$$\text{and (2):} \quad \frac{4T}{l^2} x(l-x) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi x}{l}.$$

Let's find the coefficients.

$$\begin{aligned} A_0 &= \frac{2}{\ell} \int_0^{\ell} \frac{4T}{\ell^2} x(\ell-x) dx \\ &= \frac{8T}{\ell^3} \int_0^{\ell} (\ell x - x^2) dx = \frac{8T}{\ell^3} \left[\ell \frac{x^2}{2} - \frac{x^3}{3} \right]_0^{\ell} \\ &= \frac{8T}{\ell^3} \left[\frac{\ell^3}{2} - \frac{\ell^3}{3} \right] = \frac{8T}{6} \Rightarrow A_0 = \frac{4T}{3} \end{aligned}$$

$$\begin{aligned}
 A_n &= \frac{2}{l} \int_0^l \frac{4T}{l^2} x(l-x) \cos \frac{n\pi x}{l} dx \\
 &= \frac{8T}{l^3} \left[l \int_0^l x \cos \frac{n\pi x}{l} dx - \int_0^l x^2 \cos \frac{n\pi x}{l} dx \right] \\
 &= \frac{8T}{l^3} \left[\underbrace{l x \frac{l}{n\pi} \sin \frac{n\pi x}{l}}_{=0} \Big|_0^l - \frac{l^2}{n\pi} \int_0^l \sin \frac{n\pi x}{l} dx \right. \\
 &\quad \left. - \underbrace{x^2 \frac{l}{n\pi} \sin \frac{n\pi x}{l}}_{=0} \Big|_0^l + \frac{2l}{n\pi} \int_0^l \sin \frac{n\pi x}{l} x dx \right] \\
 &= \frac{8T}{l^3} \left[+ \frac{l^2}{n\pi} \frac{l}{n\pi} \cos \frac{n\pi x}{l} \Big|_0^l - \frac{2l}{n\pi} x \frac{l}{n\pi} \cos \frac{n\pi x}{l} \Big|_0^l \right. \\
 &\quad \left. + \frac{2l^2}{n^2\pi^2} \int_0^l \cos \frac{n\pi x}{l} dx \right] \\
 &= \frac{8T}{l^3} \left[\frac{l^3}{n^2\pi^2} (\cos n\pi - 1) - \frac{2l^3}{n^2\pi^2} \cos n\pi \right. \\
 &\quad \left. + \frac{2l^2}{n^2\pi^2} \frac{l}{n\pi} \sin \frac{n\pi x}{l} \Big|_0^l \right] \quad \left. \vphantom{\frac{2l^2}{n^2\pi^2} \frac{l}{n\pi} \sin \frac{n\pi x}{l} \Big|_0^l} \right\} \begin{array}{l} \sin n\pi = 0 \\ \sin 0 = 0 \end{array} \\
 &\qquad\qquad\qquad "0 \\
 &= -\frac{8T}{l^3} \cdot \frac{l^3}{n^2\pi^2} (1 + \cos n\pi) = -\frac{8T}{n^2\pi^2} (1 + (-1)^n) \\
 &= \begin{cases} 0, & n = 2m+1 \quad (\text{odd } n) \\ -\frac{16T}{n^2\pi^2}, & n = 2m, \quad m = 1, 2, \dots \quad (\text{even } n). \end{cases}
 \end{aligned}$$

Substituting A_0 and A_n into (3), we obtain (7)
the final answer:

$$u(x,t) = \frac{2T}{3} - \frac{16T}{\pi^2} \sum_{m=1}^{\infty} \frac{\cos \frac{2m\pi x}{l}}{4m^2} e^{-\frac{4m^2\pi^2 Kt}{l^2}}.$$

(4) The displacement of the string, $u(x,t)$, obeys the wave equation: $u_{tt} - c^2 u_{xx} = 0$.

Boundary conditions: $u(0,t) = u(l,t) = 0$.

Initial conditions: $u(x,0) = \frac{4h}{l^2} x(l-x),$

$$u_t(x,0) = 0.$$

Using variable separation,

$$u(x,t) = v(x)q(t).$$

Substituting into the wave equation and dividing by $v(x)q(t)$:

$$\underbrace{\frac{1}{q(t)} \frac{d^2 q}{dt^2}}_{\text{const} = -\omega^2} - c^2 \underbrace{\frac{1}{v(x)} \frac{d^2 v}{dx^2}}_{-\omega^2} = 0.$$

The equation for $q(t)$:

$$\frac{d^2 q}{dt^2} + \omega^2 q(t) = 0.$$

General solution:

$$q(t) = A \cos \omega t + B \sin \omega t.$$

The coordinate equation:

$$\frac{d^2 v}{dx^2} + \frac{\omega^2}{c^2} v(x) = 0.$$

Introducing $k = \frac{\omega}{c}$, we have:

(8)

$$v'' + k^2 v = 0.$$

The general solution of this equation reads

$$v(x) = C_1 \cos kx + C_2 \sin kx.$$

To satisfy the boundary conditions, we require

$$v(0) = 0 \Rightarrow C_1 = 0$$

$$v(l) = 0 \Rightarrow C_2 \sin kl = 0,$$

$$\text{hence } kl = n\pi, \quad n = 1, 2, \dots$$

$$k = \frac{n\pi}{l}$$

$$\omega = kc = \frac{n\pi c}{l} \equiv \omega_n.$$

Hence, we have obtained the following solution:

$$u(x, t) = C_2 \sin \frac{n\pi x}{l} (A \cos \omega_n t + B \sin \omega_n t).$$

In fact, there is no need to keep C_2 , since A and B are arbitrary constants. Renaming them A_n and B_n , and using the superposition principle, we construct:

$$u(x, t) = \sum_{n=1}^{\infty} (A_n \cos \omega_n t + B_n \sin \omega_n t) \sin \frac{n\pi x}{l}. \quad (*)$$

To apply the initial conditions, we also need its time derivative,

$$u_t(x, t) = \sum_{n=1}^{\infty} (-A_n \omega_n \sin \omega_n t + B_n \omega_n \cos \omega_n t) \sin \frac{n\pi x}{l}.$$

Applying $u_t(x, 0) = 0$ first,

$$\sum_{n=1}^{\infty} B_n \omega_n \underbrace{\cos \omega_n t}_{=1 \text{ for } t=0} \sin \frac{n\pi x}{l} = \sum_{n=1}^{\infty} B_n \omega_n \sin \frac{n\pi x}{l} = 0 \Rightarrow B_n = 0.$$

The first initial condition then reads:

(9)

$$\frac{4h}{l^2} x(l-x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l}$$

Hence,

$$A_n = \frac{2}{l} \int_0^l \frac{4h}{l^2} x(l-x) \sin \frac{n\pi x}{l} dx$$

This integral is identical to that worked out in problem (2) (see pages 3 and 4), provided we replace T with h . Hence:

$$A_n = \begin{cases} 0, & n = 2m \\ \frac{32h}{n^3 \pi^3}, & n = 2m+1, \quad m = 0, 1, \dots \end{cases}$$

Substituting this into equation (*) on page 8, we have the final answer:

$$u(x,t) = \frac{32h}{\pi^3} \sum_{m=0}^{\infty} \frac{\sin \frac{(2m+1)\pi x}{l}}{(2m+1)^3} \cos \frac{(2m+1)\pi c}{l} t$$

- (5) Solve the problem of finding the shape of a rectangular membrane, $-a \leq x \leq a$, $-b \leq y \leq b$, under a uniform load q .

The equilibrium shape satisfies Poisson's equation

$$u_{xx} + u_{yy} = -\frac{q}{T}, \quad (1)$$

with boundary conditions

$$u(-a, y) = u(a, y) = u(x, -b) = u(x, b) = 0.$$

(a) seek solution in the form:

$$u(x, y) = u_1(x).$$

Substituting into (1), we have:

$$\frac{d^2 u_1}{dx^2} = -\frac{q}{T}.$$

Integrating:

$$\frac{du_1}{dx} = -\frac{q}{T}x + C_1,$$

(10)

and again,

$$u_1(x) = -\frac{qx^2}{2T} + C_1x + C_2.$$

Using the boundary conditions $u_1(-a) = u_1(a) = 0$:

$$-\frac{qa^2}{2T} - C_1a + C_2 = 0$$

$$-\frac{qa^2}{2T} + C_1a + C_2 = 0$$

Subtracting the 1st equation from the 2nd gives

$$2C_1a = 0 \Rightarrow C_1 = 0.$$

Then

$$C_2 = \frac{qa^2}{2T},$$

$$\text{and } \underline{u_1(x) = \frac{q}{2T}(a^2 - x^2)}.$$

$$(b) \quad \text{Let } u(x, y) = \tilde{u}(x, y) + u_1(x).$$

Substituting into (1):

$$\tilde{u}_{xx} + \tilde{u}_{yy} + \cancel{u_1''(x)} = -\cancel{\frac{q}{T}}.$$

$$\Rightarrow \tilde{u}_{xx} + \tilde{u}_{yy} = 0. \quad (2)$$

Boundary conditions:

$$\tilde{u}(-a, y) + u_1(-a) = 0 \Rightarrow \tilde{u}(-a, y) = 0. \quad (3)$$

$$\tilde{u}(a, y) + u_1(a) = 0 \Rightarrow \tilde{u}(a, y) = 0 \quad (4)$$

$$\tilde{u}(x, -b) + u_1(x) = 0 \Rightarrow \tilde{u}(x, -b) = -u_1(x) \quad (5)$$

$$\tilde{u}(x, b) + u_1(x) = 0 \Rightarrow \tilde{u}(x, b) = -u_1(x). \quad (6)$$

(c) Solving (2) by the method of variable separation, ⁽¹¹⁾
we seek solution in the form:

$$\tilde{u}(x, y) = X(x) Y(y).$$

$$Y(y) X''(x) + X(x) Y''(y) = 0 \quad : \text{divide by } XY$$

$$\underbrace{\frac{1}{X(x)} X''(x)}_{\text{depends on } x \text{ only}} + \underbrace{\frac{1}{Y(y)} Y''(y)}_{\text{depends on } y \text{ only}} = 0$$

Must be constant, $-k^2$ const, k^2 .

Equation for $X(x)$:

$$X''(x) + k^2 X(x) = 0$$

has the general solution

$$X(x) = A \cos kx + B \sin kx.$$

Using the boundary conditions at $x = -a$ and a :
(equations (3) and (4))

$$x = -a : A \cos ka - B \sin ka = 0$$

$$x = a : A \cos ka + B \sin ka = 0.$$

First adding up and then subtracting these equations,
we obtain: $A \cos ka = 0$, $B \sin ka = 0$.

Since we are looking for a symmetric (even)
solution, we choose $B = 0$

$$\text{and } \cos ka = 0 \text{ gives } ka = (n + \frac{1}{2})\pi$$

$n = 0, 1, \dots$

$$k = \frac{(n + \frac{1}{2})\pi}{a}$$

$$\Rightarrow X(x) = A \cos \frac{(n + \frac{1}{2})\pi x}{a}.$$

The equation for $Y(y)$:

(12)

$$Y''(y) - k^2 Y(y) = 0,$$

has the general solution:

$$Y(y) = C_1 e^{ky} + C_2 e^{-ky}.$$

Since we are looking for a symmetric solution ($Y(-y) = Y(y)$), we require $C_2 = C_1$, so that

$$Y(y) = C_1 (e^{ky} + e^{-ky}),$$

$$Y(y) = 2C_1 \cosh ky,$$

$$\text{or } Y(y) = 2C_1 \cosh \frac{(n+\frac{1}{2})\pi}{a} y.$$

Combining $X(x)$ and $Y(y)$, we have:

$$\tilde{u}(x, y) = A_n \cosh \frac{(n+\frac{1}{2})\pi y}{a} \cos \frac{(n+\frac{1}{2})\pi x}{a}.$$

where A_n is an arbitrary constant, and $n = 0, 1, 2, \dots$

(d) Using the superposition principle, we construct a more general solution as

$$u(x, y) = \sum_{n=0}^{\infty} A_n \cosh \frac{(n+\frac{1}{2})\pi y}{a} \cos \frac{(n+\frac{1}{2})\pi x}{a}. \quad (*)$$

It must satisfy the boundary conditions (5), (6), and since this function is even, we can apply just one of them, say for $y = b$:

$$\sum_{n=0}^{\infty} \underbrace{A_n \cosh \frac{(n+\frac{1}{2})\pi b}{a}}_{a_n} \cos \frac{(n+\frac{1}{2})\pi x}{a} = -\frac{q}{2T} (a^2 - x^2)$$

This is a Fourier-type series on $-a \leq x \leq a$,

whose coefficients, $a_n = A_n \cosh \frac{(n+\frac{1}{2})\pi b}{a}$, can be found by standard means.

$$a_n = \frac{1}{a} \int_{-a}^a \left(-\frac{q}{2T} (a^2 - x^2) \right) \cos \frac{(n+\frac{1}{2})\pi x}{a} dx$$

$$= -\frac{q}{2aT} \int_{-a}^a (a^2 - x^2) \cos \frac{(n+\frac{1}{2})\pi x}{a} dx$$

The integrand is even,
so we can change
 $\int_{-a}^a$ to $2 \int_0^a$

$$= -\frac{q}{aT} \int_0^a (a^2 - x^2) \cos \frac{(n+\frac{1}{2})\pi x}{a} dx$$

$$= -\frac{q}{aT} \left[a^2 \int_0^a \cos \frac{(n+\frac{1}{2})\pi x}{a} dx - \int_0^a x^2 \cos \frac{(n+\frac{1}{2})\pi x}{a} dx \right]$$

$$= -\frac{q}{aT} \left[a^2 \frac{a}{(n+\frac{1}{2})\pi} \sin \frac{(n+\frac{1}{2})\pi x}{a} \Big|_0^a - x^2 \frac{a}{(n+\frac{1}{2})\pi} \sin \frac{(n+\frac{1}{2})\pi x}{a} \Big|_0^a + \frac{2a}{(n+\frac{1}{2})\pi} \int_0^a \sin \frac{(n+\frac{1}{2})\pi x}{a} x dx \right]$$

$$= -\frac{q}{aT} \left[\frac{a^3 (-1)^n}{(n+\frac{1}{2})\pi} - \frac{a^3 (-1)^n}{(n+\frac{1}{2})\pi} - \frac{2ax}{(n+\frac{1}{2})\pi} \cdot \frac{a}{(n+\frac{1}{2})\pi} \cos \frac{(n+\frac{1}{2})\pi x}{a} \Big|_0^a + \frac{2a^2}{(n+\frac{1}{2})^2 \pi^2} \int_0^a \cos \frac{(n+\frac{1}{2})\pi x}{a} dx \right]$$

$\sin(n+\frac{1}{2})\pi = (-1)^n$
 $\sin 0 = 0$
 $\cos(n+\frac{1}{2})\pi = 0$

$$= -\frac{q}{aT} \cdot \frac{2a^2}{(n+\frac{1}{2})^2 \pi^2} \cdot \frac{a}{(n+\frac{1}{2})\pi} \sin \frac{(n+\frac{1}{2})\pi x}{a} \Big|_0^a$$

$$= -\frac{2qa^2}{T} \frac{(-1)^n}{(n+\frac{1}{2})^3 \pi^3} = \frac{16qa^2 (-1)^{n+1}}{T (2n+1)^3 \pi^3}$$

Writing $\frac{(n+\frac{1}{2})\pi x}{a} = \frac{(2n+1)\pi x}{2a}$, etc., we have

$$A_n = \frac{16qa^2}{T} \frac{(-1)^{n+1}}{(2n+1)^3 \pi^3} \cdot \frac{1}{\cosh \frac{(2n+1)\pi b}{2a}}$$

and substituting this into (*) on page 12, and combining with $u_1(x)$, we have for $u(x,y) = u_1(x) + \tilde{u}(x,y)$:

$$u(x,y) = \frac{qa^2}{2T} \left[1 - \frac{x^2}{a^2} + \frac{32}{\pi^3} \sum_{n=0}^{\infty} \frac{(-1)^{n+1} \cosh \frac{(2n+1)\pi y}{2a}}{(2n+1)^3 \cosh \frac{(2n+1)\pi b}{2a}} \cos \frac{(2n+1)\pi x}{2a} \right]$$